

Keywords: long video understanding • agentic AI • edge-cloud • visual routing • multimodal LLM

Caption-once, Frames-on-Demand: Visual-Need Routing for Budget-Aware Agentic Long Video Understanding

One Offline Captioning Pass, a Dual-Track Narrative Index, and a Lightweight Router Cut Frame API Calls by 71% While Improving Accuracy

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arXiv:2609.11899 • EMNLP 2026

Deploying long-video understanding on edge devices imposes two simultaneous constraints: tight compute budgets (no dense frame encoding over hours of content) and tight bandwidth budgets (no per-query frame refetch from the cloud). Prior systems sacrifice accuracy for efficiency (sparse frame methods) or efficiency for accuracy (dense MLLM methods). Caption-once, Frames-on-Demand (CFD) resolves the tradeoff: a single offline captioning pass builds a reusable dual-track narrative index, and a lightweight Visual-Need Router at query time triggers pixel retrieval only when the question requires it — 29% of the time.

AT A GLANCE

Problem: Long-video understanding on edge devices is blocked by the cost of dense frame processing and per-query frame retrieval.

Why it matters: Hours-long video QA is needed for wearables, in-car systems, and surveillance; bandwidth and compute are limited.

Method: Caption video once; build event-level story skeleton + clip-level micro-log; route each query through Visual-Need Router; retrieve frames only for perceptual sub-questions.

Result: 74.8%/60.9%/66.7% on EgoSchema/LVBench/Video-MME-Long; average 1.16 frames per query vs. 18–64 for baselines.

Evidence: Ablations on dual-track index, router design, and frame budget $R_{\max}$.

The Visual-Textual Duality

The core insight is that long-video questions separate cleanly into two types:

Structural questions (temporal order, causality, scene

transitions, narrative arc) are answered better from language memories than from dense frames. A prose description of a sequence of events conveys order and causality more compactly and reliably than 3,600 frames at 1 fps.

Perceptual questions (clothing color, on-screen text, object state, character identity) require pixels. Language captions inevitably abstract away discriminating visual details; the answer “the person is wearing a blue jacket” is useless if the question distinguishes between navy, teal, and cobalt.

Prior systems ignore this duality. Dense-frame MLLMs pay the cost of all frames for all questions. Text-only memory agents fail systematically on perceptual questions. CFD exploits the duality: build a reusable language index and retrieve frames only when pixels are genuinely needed.

The CFD Framework

Offline phase (run once at ingestion):

1. Clip segmentation. Video is divided into clips $\{c_1, \dots, c_K\}$ at scene boundaries by a local shot-detection model (no API calls).

2. Clip captioning. Each clip c_k is captioned by a compact 3B-parameter local MLLM, producing micro-log entry m_k : timestamp range, key actors/objects, described actions, verbatim on-screen text.

3. Story synthesis. A cloud MLLM writes an event-level story skeleton S (200–500 words) covering the main narrative arc, key events, and causal relationships from the full micro-log sequence.

4. Index construction. The dual-track index $\mathcal{I} = (S, \{m_k\}, \{t_k\})$ is stored locally.

Online phase (per query):

1. Story pass. Read S and identify candidate time windows $\mathcal{W}^*$ relevant to the query.

2. Micro-log pass. Read micro-logs for $\mathcal{W}^*$ and attempt a text-only answer.

3. Visual-Need Router. Classify remaining uncertainty as “perceptual” or “structural”. If structural: return text answer. If perceptual: request bounded keyframe retrieval.

4. Bounded retrieval. Request at most $R_{\max} = 4$ frames from the relevant window; integrate into the final response.

Technical Formulation

The full inference pipeline computes:

$$\hat{A} = \text{MLLM}(q, S, \{m_k : t_k \in \mathcal{W}^*\}, \{f_j : j \in \mathcal{R}(q)\})$$

where the frame retrieval set is:

$$\mathcal{R}(q) = \begin{cases} \arg \max_{|J| \leq R_{\max}} \text{Rel}(q, \{f_j\}_{j \in J}) & \phi(q, \mathcal{I}) = \text{perceptual} \\ \emptyset & \text{otherwise} \end{cases}$$

and $\phi : (q, \mathcal{I}) \rightarrow \{\text{perceptual}, \text{structural}\}$ is the Visual-Need Router.

Expected cost per query:

$$\mathbb{E}[|\mathcal{R}(q)|] = R_{\max} \cdot p_{\text{perc}} = 4 \times 0.29 = 1.16 \text{ frames}$$

vs. ~ 64 for dense-frame baselines and ~ 18 – 22 for prior agentic methods.

Visual-Need Router

The router ϕ is a 350M-parameter BERT-style encoder trained on 2,400 (question, micro-log) pairs annotated for visual necessity. It classifies a query as “perceptual” if the discriminating visual attribute cannot be recovered from the narrative text alone.

Router accuracy on 600 held-out pairs: 91.4%. Inference latency: < 1 ms on CPU — negligible vs. frame retrieval latency (200–800 ms round-trip).

Experimental Results

Method	EgoS.	LVB.	VME-L
GPT-4V-Long (dense)	72.4%	58.3%	64.1%
VideoAgent	68.9%	52.7%	59.8%
EgoVideoQA	70.3%	54.1%	61.2%
CFD (ours)	74.8%	60.9%	66.7%

EgoS. = EgoSchema; LVB. = LVBench; VME-L = Video-MME Long.

CFD improves over the best prior method by 2.4–2.8 accuracy points while using 15 – $55\times$ fewer frames per query.

Ablations and Interpretation

Without story skeleton: Accuracy drops 5.3 points on LVBench. Multi-scene temporal reasoning requires the sketch-level narrative.

Without micro-logs: Accuracy drops 3.1 points on EgoSchema. Clip-level detail is needed for fine-grained activity understanding.

Without Visual-Need Router (always retrieve): Accuracy improves by only 0.9 points but frame cost increases $3.4\times$. The router’s accuracy–efficiency tradeoff is strongly favorable.

Single-track index: Either track alone scores 4–8 points below the dual-track system, confirming the two tracks are complementary.

Reference

Weitong Cai et al. **Caption-once, Frames-on-Demand: Visual-Need Routing for Budget-Aware Agentic Long Video Understanding.** arXiv:2609.11899, EMNLP 2026. <https://arxiv.org/abs/2609.11899>